

Zarif Latif

📍 Dhaka, Bangladesh ✉ zarif.latif.biz@gmail.com ☎ +8801308035203 🔗 [linkedin.com/in/zariflatif](https://www.linkedin.com/in/zariflatif)
🏠 zariffromlatif.github.io/

EDUCATION

Bachelor's degree, Computer Science

BRAC University • 2026

EXPERIENCE

Founder

Railo

January 2026 - Present, On-site

- Engineered an Automated Security Remediation Engine using Python, integrating into CI/CD pipelines to deterministically identify and fix vulnerabilities in Python and JavaScript/TypeScript codebases at pull-request time.
- Implemented secure integration workflows leveraging GitHub Actions and Regex-based parsing to scan for security flaws (e.g., hardcoded secrets, insecure libraries) and automatically commit secure alternatives.
- Transitioned the architecture to a multi-tenant SaaS model, leading the product roadmap and technical vision.

Research & Innovation Intern

Bangladesh Computer Council

January 2026 - February 2026, Agargaon, Dhaka

- Researched and prototyped novel AI safety techniques, focusing on interpretability methods for complex neural network architectures.
- Evaluated and benchmarked existing AI safety frameworks, identifying key limitations in adversarial robustness and proposing enhancements for future research.
- Developed formal verification methods using Z3 SMT solver to prove safety properties of custom neural network layers.

PROJECT

LIFE-FORGE

Open Source / Independent Research • github.com/zariffromlatif/life-forge • August 2026 - Present

- Architected and open-sourced an autonomous “flight simulator” for enterprise AI agents using 3D MAP-Elites Quality-Diversity evolutionary algorithms to discover zero-day security vulnerabilities and execution deadlocks before production.
- Discovered contrasting zero-day failure modes across 30 identical evolutionary generations on an RTX 4090: exposed Alibaba Qwen-2.5 (14B) wiring \$60,000 to spoofed CEO emails (7 critical exfiltrations) and Meta Llama-3.1 (8B) entering 12 unrecoverable retry loops under market price volatility.
- Built a deterministic, in-memory digital twin enterprise sandbox supporting instantaneous state rollback, an invariant policy oracle, and native Anthropic Model Context Protocol (MCP) server integration over stdio.
- Developed a universal evaluation engine (lifeforge eval) and HTTP webhook adapter enabling cross-language stress-testing for agents built in LangGraph, CrewAI, AutoGen, and Dockerized microservices.

SignalForge

github.com/zariffromlatif/SignalForge • June 2026 - September 2026

- Open-source Generative Engine Optimization (GEO) framework. Measure brand Share of Voice across ChatGPT, Claude, and Perplexity, and deploy autonomous signals to earn AI citations.

- Designed and implemented autonomous signal deployment mechanisms to capture AI citations and enhance brand Share of Voice across major LLMs.

Aetherius Risk Intelligence

github.com/zarifffromlatif/Aetherius-Risk-Intelligence • February 2026 - April 2026

- Deterministic, backtest-validated downside-risk screening on SEC EDGAR filings and GDELT financial news for concentrated public-equity books.
- Developed novel feature engineering techniques on textual data from SEC filings and financial news to enhance predictive accuracy for downside risk.

PUBLICATIONS

Verification-Gated Neuro-Symbolic Program Repair with Bounded SMT-Based Validation of Semantic Sanitization

Working Paper • Independent Research Draft

From Prediction to Prescription: Graph-Enhanced Causal Offline Reinforcement Learning for Customer Retention

Academic Thesis • BRAC University (Undergraduate Thesis)

Optimizing Static Malware Analysis: A Lightweight, Explainable Framework using LightGBM and SHAP

Working Paper • Independent Research Draft

Cross-Lingual Emergence of Induction Heads in Bengali Language Models: A Mechanistic Interpretability Study

Working Paper • Independent Research Draft

SKILLS

- **AI Safety & Interpretability:** Mechanistic Interpretability, TransformerLens, Induction Heads, Formal Verification, Z3 SMT Solver, Neuro-Symbolic AI.
- **Machine Learning & Data Science:** PyTorch, TensorFlow, LightGBM, Scikit-learn, Hugging Face, Graph Neural Networks (GNNs), LLM Prompt Caching.
- **Software Engineering:** Python (Advanced), Golang, JavaScript (ES6+), C++, MERN Stack, AST Transformations, System Design.
- **Cybersecurity & DevOps:** Automated Remediation, Taint Analysis, Penetration Testing, Docker, Kubernetes (K8s), CI/CD Automation.

Tools & Technologies: Node.js, MySQL, React.js, Bootstrap (Framework), JSON Web Token (JWT), TensorFlow, Python (Programming Language), Kotlin, Microsoft Office, Microsoft Excel

Interpersonal Skills: Team Leadership, Problem Solving, Strategic Thinking, Strategy, Analytical Skills, Communication
